

AUTOMATED PODCAST TRANSCRIPTION AND TOPIC SEGMENTATION



presented by: Dharani

Problem Statement & Solution

- ▶ Problem:
- ▶ Manual transcription and analysis of podcast audio is time-consuming and error-prone.
- ▶ Solution:
- ▶ Automated system to clean audio, generate transcripts, segment topics, extract keywords, and visualize insights using a dashboard.

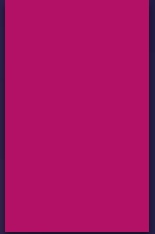
Approach

- ▶ Collect and organize raw podcast audio files
- ▶ Clean and preprocess audio for better transcription accuracy
- ▶ Convert speech to text and preprocess transcripts
- ▶ Segment text and extract important keywords
- ▶ Visualize results and display them in an interactive dashboard

Project Objectives

- ▶ - Automate podcast transcription
- ▶ - Clean raw audio files
- ▶ - Segment text into meaningful topics
- ▶ - Extract keywords
- ▶ - Visualize data using charts and word clouds
- ▶ - Provide user-friendly dashboard

System Architecture



- ▶ Audio Input
- ▶ Audio Cleaning
- ▶ Transcription
- ▶ Preprocessing
- ▶ Segmentation
- ▶ Keyword Extraction
- ▶ Visualization
- ▶ UI Dashboard

Technologies Used

- ▶ Python
- ▶ Streamlit
- ▶ Matplotlib
- ▶ WordCloud
- ▶ OS & File Handling
- ▶ NLP Libraries

- Play cleaned audio



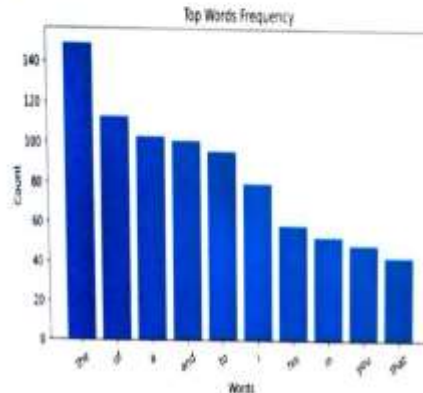
Upload Transcript Audio segments Keywords

Text Visualizations

Select transcript for visualization

adventuresholmes.01 Doyle 64kb.txt

 Top Words Frequency



Word Cloud



Limitations & Challenges

- ▶ - Large audio files take more processing time
- ▶ - Keyword extraction is basic
- ▶ - Requires manual folder setup
- ▶ - Limited scalability

Future Scope

- ▶ - Real-time transcription
- ▶ - Sentiment analysis
- ▶ - Speaker identification
- ▶ - Cloud deployment
- ▶ - Multi-language support

THANK YOU!!!